Pengfei Ma

mapengfei@mail.nwpu.edu.cn
@npuheart mfmpf1996

Research Interests

- My research centers on computational cardiology, with a primary focus on fully GPU-accelerated fluid–structure interaction for whole-heart simulations. Building on this foundation, I extend my work to peripheral vascular circulation, electromechanical coupling of cardiac tissue, and coronary perfusion modeling. Ultimately, I aim to translate these high-fidelity, high-performance simulations into clinically relevant tools that support diagnosis, treatment planning, and personalized medicine.

Education

2019 – 2025 **Ph.D.** in Mathematics (successfully defended in December 2025).
School of Mathematics and Statistics, Northwestern Polytechnical University

2018 – 2019 **M.Sc.** in Computational Mathematics (Transferred to Ph.D. Program)
School of Mathematics and Statistics, Northwestern Polytechnical University

2014 – 2018 **B.Sc.** in Information and Computing Science
School of Science, Northwestern Polytechnical University

Research Publications

Journal Articles

1. **P. Ma**, L. Cai, X. Wang, and H. Gao, "AFSI: Automated fluid-structure interaction solver development for nonlinear solid mechanics," *arXiv preprint arXiv:2509.00014*, 2025.
2. **P. Ma**, L. Cai, X. Wang, and H. Gao, "Fully GPU-accelerated, matrix-free immersed boundary method for complex fiber-reinforced hyperelastic cardiac models," *Computer Methods in Applied Mechanics and Engineering*, vol. 447, p. 118 353, 2025, ISSN: 0045-7825.
3. X. Wang, L. Cai, **P. Ma**, and H. Gao, "An efficient staggered scheme for solving the poromechanics problem of quasi-static cardiac perfusion," *International Journal for Numerical Methods in Biomedical Engineering*, vol. 41, no. 4, e70030, 2025.
4. Q. Chen, L. Cai, F. Jing, **P. Ma**, X. Luo, and H. Gao, "On the immersed boundary method with time-filter-sav for solving fluid–structure interaction problem," *Journal of Scientific Computing*, vol. 100, no. 2, p. 45, 2024.
5. Y. Liu, L. Cai, Y. Chen, **P. Ma**, and Q. Zhong, "Variable separated physics-informed neural networks based on adaptive weighted loss functions for blood flow model," *Computers & Mathematics with Applications*, vol. 153, pp. 108–122, 2024.
6. **P. Ma**, L. Cai, X. Wang, Y. Wang, X. Luo, and H. Gao, "An unconditionally stable scheme for the immersed boundary method with application in cardiac mechanics," *Physics of Fluids*, vol. 36, no. 8, 2024.
7. L. Cai, Y. Hao, **P. Ma**, G. Zhu, X. Luo, and H. Gao, "Fluid-structure interaction simulation of calcified aortic valve stenosis," *Mathematical Biosciences and Engineering*, vol. 19, no. 12, pp. 13 172–13 192, 2022.
8. L. Cai, J. Jiao, **P. Ma**, W. Xie, and Y. Wang, "Estimation of left ventricular parameters based on deep learning method," *Mathematical Biosciences and Engineering*, vol. 19, no. 7, pp. 6638–6658, 2022.

Skills

Coding	C++, Python, CUDA, MPI
Numerical	Finite Element Method, Finite Difference Method, Immersed Boundary Method

Miscellaneous Experience

Participated Projects

2019.1–2022.12	National Natural Science Foundation of China General Program (11871399), Efficient Numerical Methods and Applications for Electrophysiological Coupling in Injured Hearts
2023.1–2026.12	National Natural Science Foundation of China General Program (12271440), Multiscale and Multiphysics Coupled Modeling and Computation of Myocardial Perfusion

Software Development

AFSI	An open-source Python package for automated fluid-structure interaction modeling. `https://github.com/npuheart/afsi`
NPUHEART	A high-performance GPU-based framework for large-scale fluid-structure interaction simulations. (Not yet open-sourced)

References

Supervisors	`https://www.maths.gla.ac.uk/~xl/team.html` `https://teacher.nwpu.edu.cn/caili.html`